

BODO AGDM, AB, Canada

WMO: 712140

Lat: 52.1242N

Lon: 110.1014W

Elev: 687

StdP: 93.34

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.9	-26.9	-33.4	0.2	-29.6	-30.3	0.2	-26.8	11.6	-5.2	10.2	-6.3	3.2	300	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	30.3	17.3	28.2	16.9	26.2	16.1	19.3	26.5	18.2	25.3	17.2	23.9	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	13.1	22.1	15.7	12.1	20.8	14.6	11.3	19.7	58.2	26.5	54.3	25.2	51.0	23.7	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.5	-33.5	33.8	2.5	2.0	-35.2	35.2	-36.7	36.4	-38.1	37.5	-39.9	38.9		
(5)			-33.4	21.5	2.6	1.4	-35.2	22.5	-36.7	23.3	-38.1	24.0	-40.0	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-10.5	-10.9	-3.8	4.4	10.9	15.1	18.1	16.9	11.9	4.6	-4.1	-10.7	(6)
(7)		DBStd	12.22	8.26	7.54	7.50	5.39	4.40	3.34	3.11	3.62	4.77	5.35	6.78	7.92	(7)
(8)		HDD10.0	3151	636	584	429	178	42	3	0	1	33	179	424	642	(8)
(9)		HDD18.3	5466	894	818	687	419	231	107	42	71	198	425	674	900	(9)
(10)		CDD10.0	807	0	0	1	9	71	156	252	215	91	13	1	0	(10)
(11)		CDD18.3	79	0	0	0	0	2	10	35	27	6	0	0	0	(11)
(12)		CDH23.3	1246	0	0	1	3	64	148	478	397	149	6	0	0	(12)
(13)		CDH26.7	359	0	0	0	0	9	33	147	119	51	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.1	3.9	4.4	4.8	4.5	4.1	3.8	3.8	3.9	4.3	4.2	3.9	(14)
(15)	Precipitation	PrecAvg	385	19	11	15	28	40	70	71	52	29	17	15	15	(15)
(16)		PrecMax	522	39	23	35	81	86	120	137	127	73	41	50	28	(16)
(17)		PrecMin	242	5	4	1	2	6	23	25	6	0	2	3	4	(17)
(18)		PrecStd	77	9	5	8	18	24	27	26	26	22	9	12	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.2	15.1	22.9	28.0	30.3	32.8	32.7	31.6	24.0	13.9	7.3	(19)
(20)			MCWB	3.2	2.9	7.8	10.7	13.9	16.8	18.6	17.5	15.7	12.9	6.7	3.4	(20)
(21)		2%	DB	4.6	3.9	11.1	19.8	25.2	27.1	30.3	29.9	27.6	19.2	9.8	4.6	(21)
(22)			MCWB	1.8	1.3	5.7	9.5	12.8	16.0	18.6	17.0	14.8	10.6	4.9	1.7	(22)
(23)		5%	DB	2.8	1.7	7.9	16.5	22.7	24.7	28.1	27.6	24.1	15.9	6.7	2.5	(23)
(24)			MCWB	0.6	-0.2	3.8	7.7	12.0	15.2	17.9	16.8	13.6	8.8	3.0	0.1	(24)
(25)		10%	DB	0.8	-0.2	5.4	13.7	20.2	22.7	25.9	25.4	21.0	12.9	4.3	0.0	(25)
(26)			MCWB	-0.8	-1.6	2.4	6.4	10.8	14.2	17.3	16.0	12.3	7.1	1.6	-1.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.2	8.3	11.5	15.2	19.1	21.7	20.2	16.9	13.3	7.3	4.0	(27)
(28)			MCD B	6.0	6.2	14.5	21.8	24.8	26.2	28.8	27.0	27.0	22.6	12.6	6.8	(28)
(29)		2%	WB	2.0	1.4	5.7	9.7	13.7	17.3	19.9	18.8	15.5	11.1	5.3	1.9	(29)
(30)			MCD B	4.3	3.5	10.2	18.5	23.0	24.4	27.2	26.3	25.0	17.9	9.3	4.3	(30)
(31)		5%	WB	0.7	0.0	4.0	8.1	12.6	16.2	18.8	17.6	14.2	9.3	3.3	0.2	(31)
(32)			MCD B	2.7	1.6	7.5	15.5	21.1	23.0	25.7	25.0	22.5	14.8	6.1	2.3	(32)
(33)		10%	WB	-0.9	-1.8	2.4	6.7	11.5	15.1	17.8	16.6	13.0	7.7	1.8	-1.6	(33)
(34)			MCD B	0.8	-0.3	5.3	12.7	19.1	21.2	24.4	23.5	20.5	12.6	4.1	0.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.0	9.9	11.9	13.1	11.9	13.1	13.3	13.2	11.6	9.2	9.1	(35)
(36)			MCD BR	10.6	11.4	13.0	16.9	16.7	15.3	16.6	17.2	18.8	16.5	12.2	11.3	(36)
(37)			MCWBR	8.3	9.2	8.3	8.5	7.5	6.8	7.0	7.1	8.1	8.9	8.1	8.7	(37)
(38)		5% WB	MCD BR	10.8	11.2	12.3	15.8	14.9	13.5	14.5	14.9	17.1	15.7	11.6	11.0	(38)
(39)			MCWBR	8.8	9.2	8.2	8.4	7.2	6.8	7.0	6.9	7.8	8.8	8.0	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.284	0.293	0.341	0.369	0.374	0.368	0.381	0.325	0.287	0.261	0.241	(40)	
(41)		taud	2.338	2.292	2.423	2.326	2.294	2.330	2.381	2.313	2.441	2.526	2.457	2.366	(41)	
(42)		Ebn at Noon	771	837	911	895	882	878	878	843	861	839	765	729	(42)	
(43)		Edh at Noon	59	84	90	113	123	120	112	114	89	66	53	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.12	2.18	3.60	4.79	5.82	6.00	6.38	5.18	3.70	2.21	1.14	0.83	(44)	
(45)		RadStd	0.11	0.17	0.28	0.44	0.41	0.42	0.31	0.42	0.37	0.30	0.12	0.08	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page